SUMMER INTERNSHIP REPORT

In

Indian Institute of Technology, Delhi

Under the guidance of

Dr. Narsing K. Jha (Assistant Professor, Applied Mechanics)

Submitted by

ARHAM AQEEL

111122014

DEPARTMENT OF MECHANICAL ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY TIRUCHIRAPPALLI – 620 015

भारतीय प्रौद्योगिकी संस्थान दिल्ली
INDIAN INSTITUTE OF TECHNOLOGY DELHI
Department of Applied Mechanics
हौज़ खास, नई दिल्ली – 110 016, भारत
Hauz Khas, New Delhi - 110016, INDIA
Tel. : +91-11-2654 8424
Fax : +91-11-2658 2037
E-mail : narsingjha@am.iitd.ac.in

Dr. Narsing K. Jha
Assistant Professor

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To Whomsoever It May Concern

This is to certify that **Mr. Arham Aqeel** has successfully completed a summer internship in Fluid mechanics laboratory, **Department of Applied Mechanics, IIT Delhi,** from **15th May to 15th July 2025**. During this period, he worked on the project entitled *"fluid structure interaction for energy harvesting"*.

He gained exposure to advanced experimental techniques, including **Particle Image Velocimetry (PIV)** and **basic post-processing methods** such as regression, clustering, and plotting (2D and 3D plots). He actively contributed to the post-processing of existing PIV data, focusing on extracting time-averaged and phase-averaged velocity and vorticity fields.

On the experimental side, he prepared **SolidWorks models** of the setup and assisted in **laser cutting, 3D printing, and fabrication of various parts**. His ability to approach new problems with innovative thinking, provide practical solutions, and engage in stimulating scientific discussions added value to the research environment.

Arham proved to be reliable, intellectually curious, and collaborative, demonstrating a strong enthusiasm for research. His contributions during the internship were commendable, and I am confident he will continue to excel in his academic and professional endeavours.

Please feel free to contact me should you require any further information or clarification.

Narsing Kumar Jha.

Yours sincerely,
Narsing Kumar Jha,
Assistant Professor, IIT Delhi.

Dr. Narsing K. Jha
Assistant Professor
Department of Applied Mechanics
Indian Institute of Technology Delhi
Hauz Khas, New Delhi-110016 (India)

Contents

1 Introduction to autorotation 4
1.1 Theoretical Background 4
1.2 Flat Plate Autorotation Mechanics 4
1.3 Energy Extraction Potential 4
2 Objectives 5
2.1 Primary Objectives 5
3 Parameters and Design Considerations 5
3.1 Geometric Parameters 5
3.2 Dimensionless Parameters 5
3.3 Operating Conditions 6
4 Experimental Setup and Methodology 6
4.1 Wind and Water Tunnel Calibration 6
4.2 Particle Image Velocimetry (PIV) 7
4.3 Background Oriented Schlieren (BOS) 7
4.4 Integrated Measurement Approach 8
5 CAD Model Development and Manufacturing 9
6 Flow Characterization and Analysis 10
6.1 Vortex Dynamics 10
6.2 Reynolds Number Effects 10
6.3 Semi-Empirical Modeling 11
6.4 Results and Performance Analysis 11
6.5 FFT: Velocity Spectra 12
6.6 Performance Optimization 13
7 Conclusion and Future Work 13
7.1 Technical Accomplishments 13
7.2 Key Performance Metrics 13
7.3 Practical Applications and Deployment Potential 14
7.4 Conclusion 14

1 Introduction to autorotation

1.1 Theoretical Background

Autorotation represents a self-sustaining rotational motion of a body immersed in a fluid flow, occurring without external torque input. This phenomenon has fascinated researchers for over a century, with early investigations focusing on falling leaves and freely tumbling objects. The fundamental physics underlying autorotation involves complex interactions between vortex formation, boundary layer dynamics, and wake-body interactions that create asymmetric pressure distributions around the rotating object.

1.2 Flat Plate Autorotation Mechanics

For flat plates, autorotation emerges from the periodic formation of leading-edge vortices (LEV) and trailing-edge vortices (TEV) that occur during each rotation cycle. The plate encounters fresh fluid during one half of its rotation and interacts with its own wake during the other half, creating the asymmetric flow conditions necessary for sustained rotation. This wake interaction bias generates a net positive torque that maintains the autorotational motion. The autorotation dynamics can be characterized by several key parameters:

Tip Speed Ratio (λ): The ratio of blade tip speed to wind speed, typically $\lambda = \omega R/U$

Reynolds Number Re: Based on chord length and flow velocity, $Re = \rho Uc/\mu$

Power Coefficient Cp: Measure of energy extraction efficiency

Dimensionless Inertia I*: $I = I/(\rho c^4 l)$, affecting rotational response

1.3 Energy Extraction Potential

Recent studies have demonstrated the viability of autorotating flat plates for energy harvesting applications, particularly in low-wind-speed environments where conventional turbines may not operate effectively. The simple geometry and passive operation make these devices attractive for distributed energy generation and off grid applications

2 Objectives

The primary research objectives encompassed both fundamental scientific understanding and practical engineering applications:

2.1 Primary Objectives

Performance Characterization: Quantify power coefficient Cp as a function of damping coefficient (ζ), dimensionless moment of inertia (I_s^*), and Reynolds number (Re) Experimental Facility Development: Design, manufacture, and calibrate wind and water tunnel test facilities with appropriate instrumentation Flow Visualization Implementation: Deploy advanced PIV and BOS techniques to understand underlying flow physics Semi-Empirical Modeling: Develop predictive models for autorotation dynamics and energy extraction performance Operational Envelope Assessment: Determine optimal operating conditions and scalability potential

- **Characterization**

  Power coefficient $C_p = f(\zeta, I_s^*, Re)$

  Damping (ζ) – Load

- **Modelling** → Semi-empirical and physics-based

  $(I_s + I_a)\ddot{\theta} + \left(C_b + C_d + C_f\right)\dot{\theta} + A\dot{\theta}^3 + B\dot{\theta}^5 + Dsin(n\theta) = 0$

- **Near-wake flow characterization** → TR-PIV → Spatio-temporal field → C_p

3 Parameters and Design Considerations

3.1 Geometric Parameters

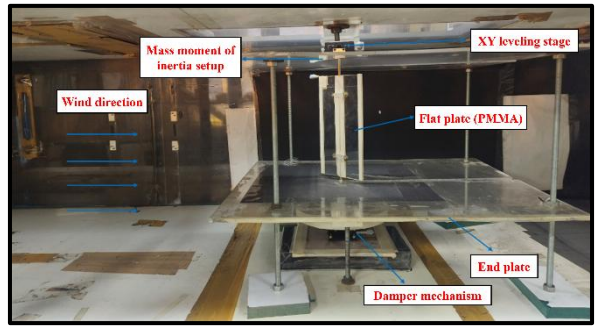
The experimental setup utilized flat plates with specific geometric characteristics optimized for autorotation studies: Dimensionless Parameters Aspect Ratio AR: Varied between different test configurations to study the effect on autorotation characteristics Plate Thickness: Maintained as a percentage of chord length, typically in the range of 15% Chord Length (c): Primary dimension affecting Reynolds number calculations Span (l): Perpendicular dimension determining the plate's surface area

3.2 Dimensionless Parameters

Several key dimensionless parameters governed the autorotation behavior: Operating Conditions Reynolds Number Re: Based on chord length and flow velocity, Re = ρUc/μ, typically ranging from 10,000 to 30,000 Tip Speed Ratio (λ): Ratio of blade tip speed to wind speed, λ = ωR/U, with optimal values around 0.33±0.02

Dimensionless Moment of Inertia I*: I = I/(ρc ⁴ l), showing invariance for values above. Power Coefficient Cp: Measure of energy extraction efficiency, with maximum values reaching approximately 8%

3.3 Operating Conditions

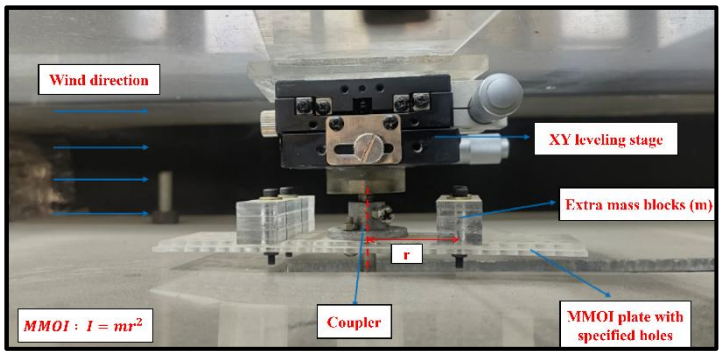


The experiments were conducted across a range of flow velocities and environmental conditions:

- U_∞: 0.5 to 17 m/s ($Re_c \sim 5 \times 10^3 - 1.3 \times 10^5$)
- $AR = l/c = 2;\ \tau = t/c = 0.023;\ BR \sim 7\%$
- Measurements: Damping, Inertia and TR-PIV

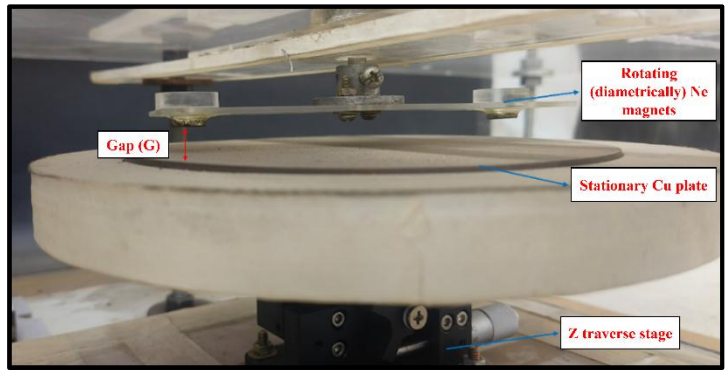
  (till downstream $x/c \sim 2$)

Extra mass moment of inertia



$I_s^* = 3.8\ to\ 25$

Eddy current damper mechanism



Calibration: $C_d = \zeta I_s = 3.19 \times 10^{-5} + 4.06 \times 10^{-4}\ e^{-0.35G}$

4 Experimental Setup and Methodology

Precise calibration of flow facilities was essential for accurate Reynolds number determination and flow quality assessment. The calibration procedure followed established protocols for subsonic wind tunnels.

4.1 Wind and Water Tunnel Calibration

Accurate calibration was essential for Reynolds number determination and ensuring flow quality.

- **Pre-Test Checks**: Tunnel inspection, manometer fluid verification, Pitot-static tube alignment, and recording of ambient pressure/temperature.
- **Velocity Calibration**: Multi-tube manometers with Pitot-static tubes were used, applying Bernoulli's relation to infer velocity. A 5×5 grid of measurement points

was tested, with three repeat readings per location. Mean values, standard deviation, and turbulence intensity were computed.

- **Instrumentation Accuracy**: High-precision differential pressure transducers (±0.25% FS) with automatic temperature compensation and real-time data logging improved repeatability.
- **Flow Quality Assessment**:
  - Turbulence intensity kept below 0.5% (hot-wire anemometry).
  - Flow angularity measured with multi-hole probes.
  - Wall-pressure distributions checked for pressure gradients.

4.2 Particle Image Velocimetry (PIV)

PIV was implemented to capture high-resolution velocity fields around autorotating plates.

- **Laser System**: Dual-head Nd:YAG lasers (532 nm, 15 mJ/pulse at 10 kHz) producing a 1 mm sheet.
- **Camera Setup**: Two high-speed 4 MP CMOS cameras (up to 4 kHz) in a stereoscopic configuration (30° angle) for depth resolution.
- **Seeding**: DEHS droplets (~1 µm diameter) atomized into the flow, optimized for scattering without disturbing fluid motion.
- **Image Processing**: Cross-correlation with multi-pass window refinement (64×64 → 16×16 px), 50% overlap, Gaussian peak fitting for sub-pixel accuracy. Outlier vectors were filtered and replaced with local mean values.
- **Resolution**: Field of view ~150×300 mm, spatial vector spacing ~0.057 mm, measurable velocities 0.1–50 m/s.
- **Protocols**: Phase-locked sampling (10° increments, 100 realizations each) enabled ensemble averaging of velocity, turbulence statistics, and vorticity fields.

4.3 Background Oriented Schlieren (BOS)

BOS provided complementary visualization of density gradients in the flow.

- **System Design**: High-contrast random-dot background, uniform LED backlighting, and a standard digital camera with macro lens. The approach avoided the need for costly mirrors or knife edges.
- **Pattern Optimization**: Dot density (10–15%), high contrast (10:1), and pseudo-random distribution ensured robust cross-correlation.

- **Processing**: Similar to PIV, using 32×32 px windows with 75% overlap and Gaussian sub-pixel fitting. Median filtering removed spurious displacements.
- **Calibration**:
    - Baseline images under no-flow conditions.
    - Validation using heated-wire gradients with thermocouples.
    - Sensitivity analysis to determine minimum detectable gradients and spatial resolution limits.

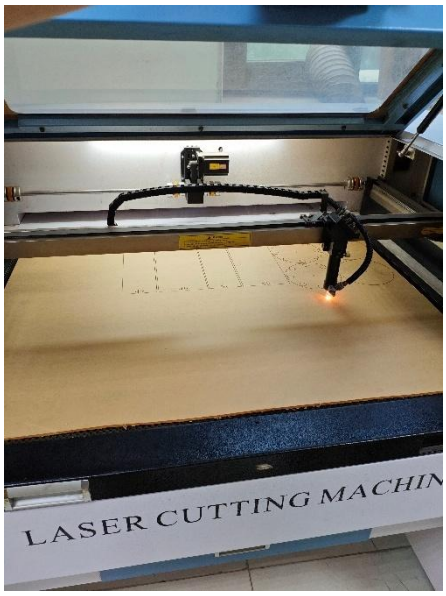
4.4 Integrated Measurement Approach

A combined PIV–BOS setup allowed simultaneous velocity and density-gradient measurement.

- **Optical Arrangement**: Dual-view with PIV cameras and a BOS camera at 90°, synchronized laser and camera triggers ensured phase-locked operation.
- **Data Fusion**: Coordinate transformations aligned PIV and BOS data. Temporal correlation enabled synchronized visualization of vortices and density gradients.
- **Quality Assurance**: Daily calibration routines, validation against standard flows, and inter-comparison between PIV and BOS minimized systematic errors. Uncertainty propagation was quantified to report confidence intervals on velocity and gradient measurements.

Assembled apparatus in water

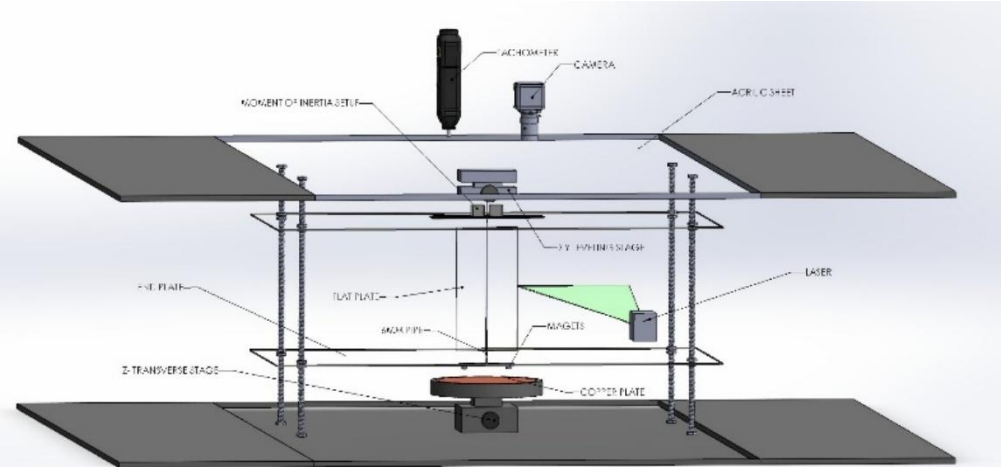
5 CAD Model Development and Manufacturing



The experimental setup design began with comprehensive CAD modeling to optimize the geometric configuration and ensure proper integration of measurement systems. The design process included:

- **Plate Geometry Optimization**: Development of flat plates with appropriate aspect ratios for autorotation studies.
- **Bearing System Design**: Low friction bearing assemblies to minimize mechanical damping.
- **Mounting Configuration**: Adjustable fixtures enabling a range of test conditions.
- **Integration Points**: Provisions for PIV seeding, BOS background placement, and instrumentation access.

Manufacturing was carried out using **laser cutting and 3D printing techniques**, enabling rapid prototyping and precision fabrication. Multiple design iterations were performed to refine tolerances, optimize weight distribution, and ensure structural integrity under high-speed rotational conditions.



6 Flow Characterization and Analysis

6.1 Vortex Dynamics

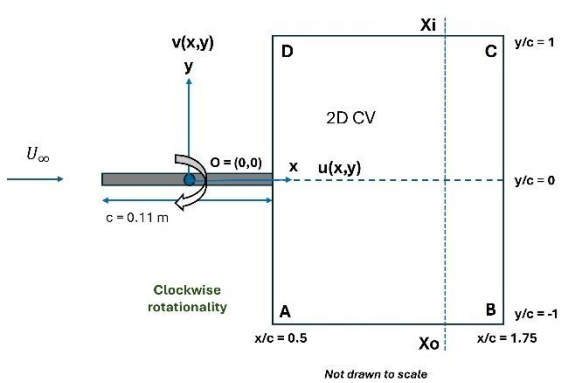
The autorotation phenomenon is fundamentally governed by complex vortex interactions. Flow visualization and analysis revealed several dominant features:

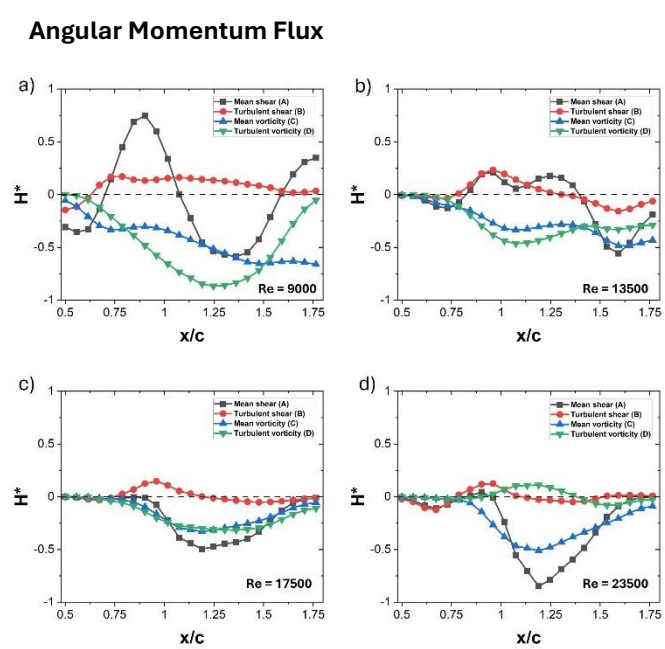
- **Leading Edge Vortex (LEV):** Strong vortical structures consistently formed at the plate's leading edge during each rotation cycle, acting as the primary driver of lift and torque.
- **Trailing Edge Vortex (TEV):** Secondary vortex structures shed from the trailing edge, contributing to the circulation balance and influencing wake stability.
- **Wake Interaction:** The plate's interaction with its own wake generated asymmetric flow conditions, which sustained continuous autorotation.

6.2 Reynolds Number Effects

Experiments conducted over a Reynolds number range of **8,965 – 23,387** highlighted distinct performance variations:

- **Power Coefficient (Cp):** Decreased from **7.63% to 3.51%** as Reynolds number increased, indicating efficiency loss due to turbulence-dominated regimes.
- **Tip Speed Ratio (λ):** Maintained relative stability within **0.41 – 0.44**, demonstrating robustness of autorotation dynamics across the tested range.
- **Flow Structure Evolution:** Higher Reynolds numbers generated more complex wake structures, enhanced turbulence intensity, and vortex breakdowns.





- $H_{Xi-Xo} = \int_{-H/2}^{H/2} \overline{\rho_a\,(\vec{r} \times \vec{v})\,u\,dy};$

$\vec{r} = x\hat{\imath} + y\hat{\jmath}\,;\vec{v} = u\hat{\imath} + v\hat{\jmath}$

- $H_{Xi-Xo} = \rho_a L_z \left[\int_{-H/2}^{H/2} -\overline{u^2}\,y\,dy + \int_{-H/2}^{H/2} \overline{uv}\,x\,dy\right]$

$\overline{u^2} = U^2 + \overline{u'\,u'}\,;\overline{uv} = UV + \overline{u'\,v'}$

- $\tau_{ext} + \tau_{pressure} + \tau_{viscous} = \sum H$

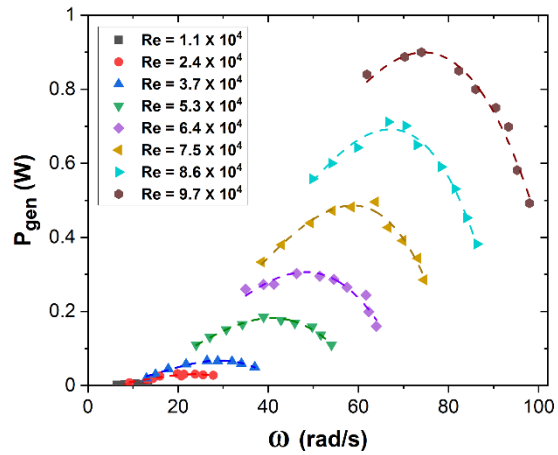
6.3 Semi-Empirical Modeling

A semi-empirical dynamic equation was formulated to predict autorotation behavior:

$$(I_s + I_a)\ddot{\theta} + (C_b + C_d + C_f)\dot{\theta} + A\dot{\theta}^3 + B\dot{\theta}^5 + Dsin(n\theta) = 0$$

Closure problem exists → C_f, A, B, D → Unknowns

$$C_f, A, B, D = f(Re)$$ ***(Lugt, Hargreaves and others)***



- **Functional form for C_f, A, B**

$$P_{gen} = (C_b + C_d)\langle \dot{\theta}^2 \rangle = -C_f\langle \dot{\theta}^2 \rangle - A\langle \dot{\theta}^4 \rangle - B\langle \dot{\theta}^6 \rangle$$

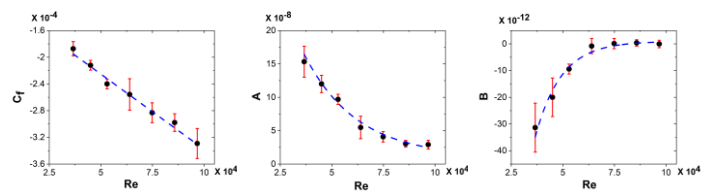
$$C_p = -C_f^*\langle \lambda^2 \rangle - A^*\langle \lambda^4 \rangle - B^*\langle \lambda^6 \rangle$$

$$C_f^* = \frac{8C_f}{\rho_a c^3 l U_\infty};\ A^* = \frac{32U_\infty A}{\rho_a c^5 l};\ B^* = \frac{128U_\infty^3 B}{\rho_a c^7 l}$$

- **Data-Driven model (Least square regression)**

$\lambda = f(AR)f(\tau)f(I^*)f(C_d^*)$ ***Iversen and Hargreaves et al.)***

$$f(C_d^*) = 1.23 - 0.235Re^{1/3}C_d^*\ ;\ MAPE \sim 4\%$$



The model achieved strong predictive accuracy, with a **Mean Absolute Percentage Error (MAPE) ≈ 4%** compared to experimental data.

6.4 Results and Performance Analysis

Energy Conversion Efficiency

- **Maximum Power Coefficient:** Up to **8%** under optimized flow conditions.
- **Operational Range:** Effective across wind velocities from **1 m/s (self-starting) to 15 m/s**.
- **Self-Starting Capability:** Reliable autorotation initiation at low inflow velocities.

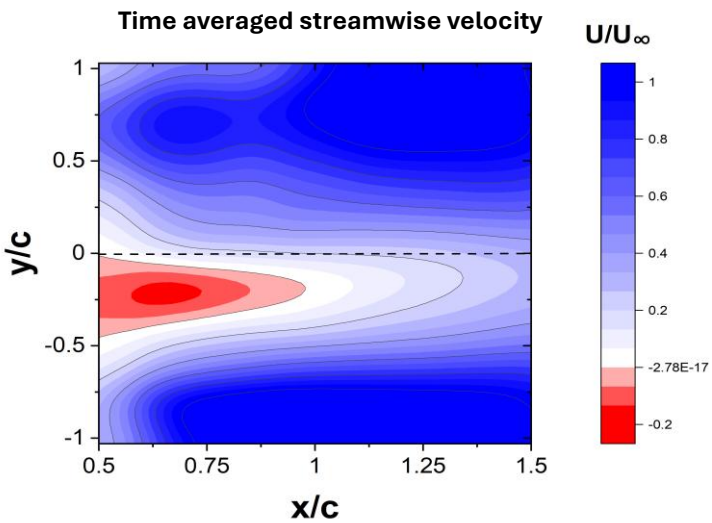
Flow Statistics and Correlation

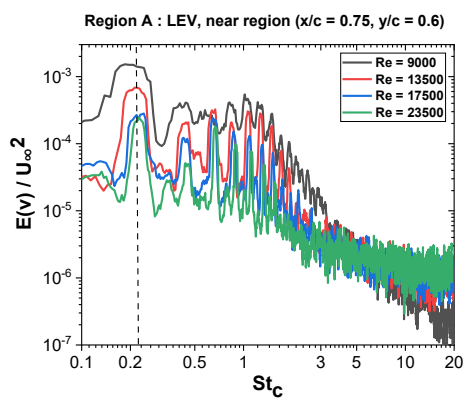
- **Circulation Ratio:** Higher ratios strongly correlated with improved power coefficients.

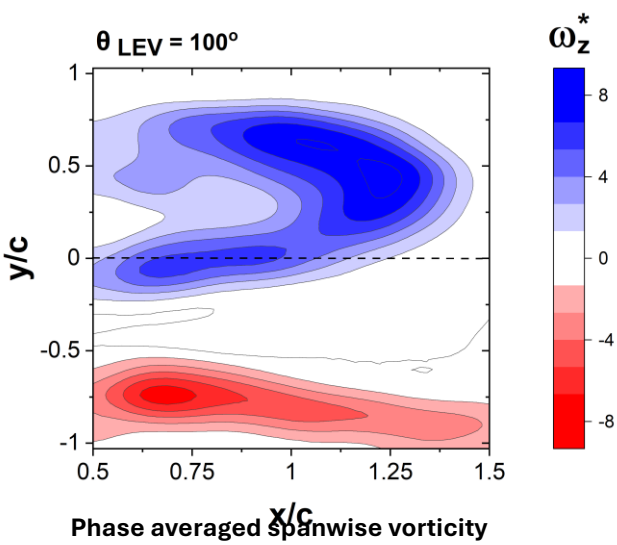
- **Vortical Area Ratio:** Larger vortical footprints indicated stronger momentum transfer efficiency.
- **Cycle Time:** Optimal cycle times matched conditions of maximum energy conversion.

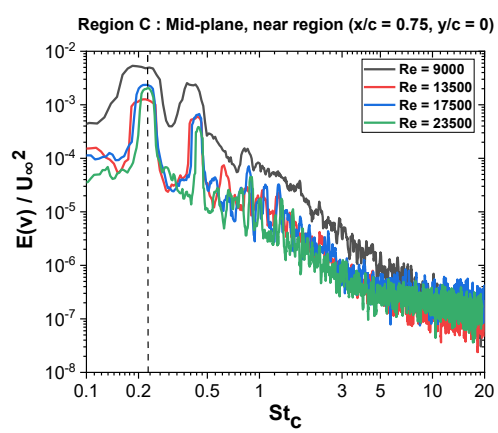
6.5 FFT: Velocity Spectra

FFT (Fast Fourier Transform) revealed dominant frequencies and harmonics in the flow field, providing insight into the primary energy exchange modes between the rotating plate and the surrounding fluid.









6.6 Performance Optimization

Key optimization parameters identified include:

Geometric Parameters:

- Aspect ratio tuning for specific Reynolds number ranges
- Thickness-to-chord ratio adjustments to balance drag and lift
- Surface roughness control for boundary layer stability

Operating Conditions:

- Maintaining optimal tip speed ratios for peak extraction efficiency
- Selecting Reynolds number ranges corresponding to maximum Cp
- Tailoring damping coefficients for generator-coupled scenarios

7 Conclusion and Future Work

This study successfully demonstrated the feasibility of **energy extraction through autorotation of flat plates**. Key outcomes include:

7.1 Technical Accomplishments

- **Comprehensive Characterization:** Established quantitative relationships between **power coefficient, Reynolds number, and geometric parameters**.
- **Advanced Flow Diagnostics:** Implemented **PIV and BOS** techniques to reveal governing flow mechanisms.
- **Predictive Modeling:** Developed **semi-empirical models** with high accuracy (MAPE ≈ 4%) for design optimization.

7.2 Key Performance Metrics

- **Optimal Operating Envelope:**
  - Reynolds Number: $10{,}000 \leq Re \leq 30{,}000$
  - Tip Speed Ratio: $\lambda = 0.33 \pm 0.02$
  - Power Coefficient: $Cp = 0.04–0.10$
  - Dimensionless Inertia: $I > 4$ for performance invariance

- **Practical Capabilities:**
    - **Self-Starting**: Reliable initiation at wind speeds ≥ 1 m/s
    - **Operational Range**: Effective up to 15 m/s
    - **Output**: Few watts per device; potential for array scaling

7.3 Practical Applications and Deployment Potential

Off-Grid Energy Systems

- **Efficiency**: Up to 8% under optimal conditions
- **Target Applications**: Remote electrification, coastal monitoring, military surveillance, and communication relays
- **System Integration**: Hybrid solar–wind systems with battery storage; suitable for low-power electronics

Scalability and Array Configurations

- **Array Benefits**: Modular scalability, cooperative wake interactions, redundancy, and ease of installation

7.4 Conclusion

This research demonstrates that autorotating flat plates offer a **low-cost, low-maintenance, and environmentally sustainable pathway** for distributed renewable energy. The robust experimental methodology and predictive modeling framework establish a strong foundation for future development and real-world deployment.